

Skin and Eye

Outcomes of choroidal melanomas treated with eye physics plaques: A 25-year review

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ABSTRACT

PURPOSE: To review long-term outcomes of the University of Southern California Plaque Simulator (PS) software and Eye Physics (EP) plaques. We hypothesize that the PS/EP system delivers lower doses to critical ocular structures, resulting in lower rates of radiation toxicity and favorable visual outcomes compared to Collaborative Ocular Melanoma Study plaques, while maintaining adequate local tumor control.

METHODS AND MATERIALS: Retrospective review of 133 patients treated for choroidal melanoma with ¹²⁵I brachytherapy, using PS software and EP plaques, from 1990 through 2015. A dose of 85 Gy at a rate of 0.6 Gy/h was prescribed to the tumor apex (with a typical margin of 2 mm) over 7 days. Primary outcomes were local tumor recurrence, globe salvage, and metastasis. Secondary outcomes were changes in visual acuity and radiation complications.

RESULTS: With median followup of 42 months, 5-year Kaplan–Meier estimated rates for tumor control, globe salvage, and metastatic-free survival were 98.3%, 96.4%, and 88.2%, respectively. Median doses to the macula and optic nerve were 39.9 Gy and 30.0 Gy, respectively. Forty-three percent of patients developed radiation retinopathy, and 20% developed optic neuropathy; 39% lost ≥ 6 Snellen lines of vision.

CONCLUSIONS: The PS/EP system is designed to improve the accuracy and conformality of the radiation dose, creating a steep dose gradient outside the melanoma to decrease radiation to surrounding ocular structures. We report favorable rates of local tumor control, globe salvage, metastases, and radiation complications when compared to the Collaborative Ocular Melanoma Study and other studies. Overall, the PS/EP system results in excellent tumor control and appears to optimize long-term visual and radiation-related outcomes after brachytherapy. © 2018 Elsevier Inc. All rights reserved.

Keywords:

Uveal melanoma; Choroidal melanoma; Brachytherapy; Eye physics; Plaque simulator; Radiation retinopathy

Received 14 April 2018; received in revised form 3 July 2018; accepted 6 July 2018.

Financial disclosure: Financial support was provided by an unrestricted departmental grant from Research to Prevent Blindness. Dr. Astrahan holds an ownership position in Eye Physics LLC, which was incorporated in 2007 to continue development of the Plaque Simulator software and Eye Physics plaques following Dr. Astrahan's emeritus retirement from the University of Southern California (USC) in 2010. During 1990–2010, no outside funding for development or material support for any of his contributions was received by USC. No compensation was received for any patient in this study. From 1995 to 2015, USC and Dr. Astrahan shared a royalty derived from licensed distribution of the Plaque Simulator software to other institutions. The authors have no other disclosures to report.

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